

Example 7: Translate the statement: "Every real number except zero has a multiplicative inverse." (A multiplicative inverse of a real number x is a real number y such that $xy=1$.)

Soln: Domain: The set of real numbers.

$$\forall x (x \neq 0) \Rightarrow \exists y (xy=1)$$

Example 8: Use quantifiers to express the definition of the limit of a real valued function $f(x)$ of a real variable x at a point a in its domain.

Soln: Recall that the definition of the statement $\lim_{x \rightarrow a} f(x) = L$ is: For every real number $\epsilon > 0$ there exists a real number $\delta > 0$ such that $|f(x) - L| < \epsilon$ whenever $0 < |x - a| < \delta$. This definition of a limit can be phrased in terms of quantifiers by

$$\forall \epsilon \exists \delta \forall x (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon), \text{ (Oshadharan)}$$

where the domain for the variables δ and ϵ consists of all positive real numbers and for x consists of all real numbers.

This definition can also be expressed as $\forall \epsilon > 0 \exists \delta > 0 \forall x (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon)$ where the domain for the variables ϵ and δ consists of all real numbers, rather than just the positive real numbers.

1.5.5 Translating from Nested Quantifiers to English

Expressions with nested quantifiers expressing statements in English can be quite complicated. The first step in translating such an expression is to write out what the quantifiers and the predicates in the expression mean. The next step is to express this meaning in a simpler sentence.

Example 9: Translate the statement $\forall x (C(x) \vee \exists y (C(y) \wedge F(x, y)))$ to English, where $C(x)$ is " x has a computer", $F(xy)$ is " x and y are friends," and the domain for both x and y consists of all students in your school.